

Classification Lecture 6

Wednesday, 2 October 2024

10:07 AM

$\hat{c}_1, \hat{c}_2 \dots \hat{c}_k \leftarrow$ classes (y & $\hat{y}$ will take these values)

$\vec{x}_i$: input (p -dimensional)

n data points for training

Find $\hat{y}$ for a new data point $x = (x_1', x_2' \dots x_p')$

Bayesian Principles

Bayes' Theorem: $P(C_k | x_1=x_1', x_2=x_2' \dots, x_p=x_p')$

$$P(C_k | x) = \frac{P(x | C_k) \cdot P(C_k)}{P(x)}$$

Diagram illustrating the components of Bayes' Theorem:

- $P(C_k | x)$ is labeled **Posterior probability**.
- $P(x | C_k)$ is labeled **likelihood**.
- $P(C_k)$ is labeled **Prior Probability**.
- $P(x)$ is labeled **Evidence**.

eg. Let's consider a spam filtering dataset where we have only two classes. Spam(S) and Not Spam(N). Assume we know the following probabilities:

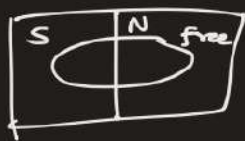
- Prior Probabilities : $P(S) = 0.2$, $P(N) = 0.8$
(based on historical data)

- Likelihoods : $P(\text{"free"} | S) = 0.5$, $P(\text{"free"} | N) = 0.1$
(based on occurrence of word "free" in historical data)

Given a new email which contains the word "free", will you classify it as a Spam email?

Sol.

$$P(S | \text{"free"}) = \frac{P(\text{"free"} | S) \cdot P(S)}{P(\text{"free"})}$$



$$P(\text{"free"}) = P(\text{"free"}, S) + P(\text{"free"}, N)$$

$$= P(\text{"free"} | S) \cdot P(S) + P(\text{"free"} | N) \cdot P(N)$$

$$= 0.5 \times 0.2 + 0.1 \times 0.8$$

$$= 0.1 + 0.08 = 0.18$$

$$P(S | \text{"free"}) = \frac{0.5 \times 0.2}{0.18} = \frac{0.1}{0.18} \approx 0.56$$

$$P(N | \text{"free"}) = \frac{0.1 \times 0.8}{0.18} = \frac{0.08}{0.18} \approx 0.44$$

Yes, we will classify the email as spam

> Do you really need to calculate $P(\text{"free"})$?

$$P(S|\text{"free"}) = \frac{P(\text{"free"}|S) \cdot P(S)}{P(\text{"free"})} \quad P(N|\text{"free"}) = \frac{P(\text{"free"}|N) \cdot P(N)}{P(\text{"free"})}$$

$$P(S|\text{"free"}) \propto P(\text{"free"}|S) \cdot \underline{P(S)} \quad \checkmark$$

$$P(N|\text{"free"}) \propto P(\text{"free"}|N) \cdot \underline{P(N)} \quad \checkmark$$

What happens in higher dimensions?

	x_1 (A/B/C)	x_2 (X/Y)	x_3 (P,Q,R)	y (0/1)
1	A	Y	Q	1
2	⋮	⋮	⋮	⋮
10	B	X	R	0

← training dataset

$K=2, P=3, N=10$
 $x_1: 3$
 $x_2: 2$
 $x_3: 3$ different values

Figure out the class of new data point $(x_1=B, x_2=X, x_3=R)$
 x

$$P(O|x) = \frac{P(x|O) \cdot P(O)}{P(x)} \Rightarrow P(O|x) \propto P(x|O) \cdot P(O)$$

$$P(O|x=[B,x,R]) \propto \underbrace{P(x=[B,x,R]|O)}_{\substack{\downarrow \\ ?}} \cdot \underbrace{P(O)}_{\substack{\downarrow \\ \text{easy to} \\ \text{calculate from} \\ \text{dataset}}}$$

$$P(x=[B,x,R]|O)$$

$$= P(x_1=B, x_2=x, x_3=R|O)$$

$$= \underline{P(x_1=B|O)} \cdot \underline{P(x_2=x|O)} \cdot \underline{P(x_3=R|O)}$$

** (Assuming conditional independence of input features given the label)

$$3 \times 2 \times 3 = \underline{18}$$

	x_1 (A/B/C)	x_2 (X/Y)	x_3 (P/Q/R)	y (0/1)
①	A	X	P	0 ←
②	A	Y	Q	1 ←
③	B	Y	Q	1 ←
④	C	X	R ✓	0 ←
⑤	C	Y	P	1 ←
⑥	B	Y	Q	0 ←
⑦	A	X	R	1 ←

$$x = [B, x, R]$$

$$x_1 = B, x_2 = X, x_3 = R$$

$$P(y=0) = 3/7$$

$$P(y=1) = 4/7$$

$$P(x = [\dots]) \\ P(x|y=0)$$

$$P(y=0 | x_1=B, x_2=X, x_3=R)$$

$$\propto \underbrace{P(x_1=B, x_2=X, x_3=R | y=0)}_{\text{conditional independence assumption}} \cdot \overset{3/7}{\underline{P(y=0)}}$$

$$\propto P(x_1=B|y=0) \cdot \underline{P(x_2=X|y=0)} \cdot P(x_3=R|y=0) \cdot P(y=0)$$

$$= \frac{1}{3} \times \frac{2}{3} \times \frac{1}{3} \times \frac{3}{7} = \frac{2}{21}$$

$$P(y=1 | x_1=B, x_2=x, x_3=R)$$

$$\propto P(x_1=B | y=1) \cdot P(x_2=x | y=1) \cdot P(x_3=R | y=1) \cdot P(y=1)$$

$$= \frac{1}{4} \times \frac{1}{4} \times \frac{1}{4} \times \frac{4}{7} = \frac{1/16}{7}$$

$$P(y=0 | x_1=...) > P(y=1 | x_1=...)$$

class ①

The "Naïve" Assumption:

The fundamental Naïve Bayes assumption is that all predictors (features) are conditionally independent given the class label. This assumption simplifies the calculation of likelihood $P(x|C_k)$ by allowing it to be expressed as the product of the probabilities of individual predictors.

$$P(x|C_k) = P(x_1, x_2, \dots, x_p | C_k) = P(x_1 | C_k) \cdot P(x_2 | C_k) \dots P(x_p | C_k)$$

$$P(x_1, x_2, \dots, x_p | C_k) = \prod_{i=1}^p P(x_i | C_k)$$

Advantages

- Computational Simplicity
- Model training & Scaling
- Performance in the presence of irrelevant features.

Disadvantages

- Feature correlation
- Data scarcity & zero probability problem
- Bias towards more frequent classes
 - ↳ $P(C_k | x) \propto P(C_k)$

Number of parameters in Naive Bayes classifier & the impact of the Naive Bayes assumption

$C_1, C_2 \dots C_k \leftarrow$ classes (y & $\hat{y}$ will take these values)

$\vec{x}_i$: input (p -dimensional)

n data points for training

Find $\hat{y}$ for a new data point $x = (x_1, x_2 \dots x_p)$

k classes

n data points

p features / predictors / attributes

$\hookrightarrow$ every feature can take ' m ' different values

$$P(C_k | x) \propto P(x | C_k) \cdot P(C_k)$$

How many different possibilities for $\vec{x}$?

$$\frac{m}{x_1} \times \frac{m}{x_2} \times \dots \times \frac{m}{x_p} = m^p$$

• Using Bayesian classifier

$$P(C_k | x) \propto \underbrace{P(x | C_k)}_{\text{likelihood}} \cdot \underbrace{P(C_k)}_{\text{prior}}$$

For a particular value C_k ,

How many parameters

we need to compute? (For likelihood)

$$\hookrightarrow m^P - 1$$

We need to do this for K classes.

$$\# \text{ parameters (For likelihood)} = K(m^P - 1)$$

For prior:- $P(C_1), P(C_2) \dots P(C_K)$

$$\therefore \# \text{ parameter (For prior)} = K - 1$$

$$P(C_k) = 1 - (P(C_1) + P(C_2) + \dots + P(C_{K-1}))$$

x_1	x_2	y
A/B	C/D	0/1

- ① $P(x_1 = A, x_2 = C | \underline{y=0})$
 - ② $P(x_1 = B, x_2 = C | \underline{y=0})$
 - ③ $P(x_1 = A, x_2 = D | \underline{y=0})$
 - ④ $P(x_1 = B, x_2 = D | \underline{y=0})$
- $y=1, \dots$

$$\textcircled{4} = 1 - \textcircled{1} - \textcircled{2} - \textcircled{3}$$

$\therefore$ Total parameters using Bayesian classifier

$$= \underbrace{k(m^p - 1)}_{\text{likelihood}} + \underbrace{(k-1)}_{\text{prior}} = \boxed{k \cdot m^p - 1}$$

eg: 10 features, Each feature can take 5 different values,
3 classes. How many parameters without Naïve assumption?

Sol: $3(5^{10} - 1) + (3 - 1)$

$$= 3 \times 5^{10} - 1 \approx 30 \text{ million}$$

• Using Naive Bayes' classifier

$$P(C_k | x) \propto \underbrace{P(x_1 | C_k)}_{\text{likelihood}} \cdot \underbrace{P(C_k)}_{\text{prior}}$$

$$= \underbrace{P(x_1 | C_k)}_{\substack{\text{m different} \\ \text{values}}} \cdot \underline{P(x_2 | C_k)} \cdots \underline{P(x_p | C_k)} \cdot \underline{P(C_k)} \quad (\text{Using Naive Bayes assumption})$$

(m-1) parameters

$\left\{ \begin{array}{l} P(x_1=1 | C_k) \\ P(x_1=2 | C_k) \\ \vdots \\ P(x_1=m | C_k) \end{array} \right\}$ sum to 1
deduce this using the remaining m-1 values

parameters for likelihood = $p(m-1)$ for one class

For k classes = $k \cdot p \cdot (m-1)$

parameters for prior = $(k-1)$

$$\therefore \text{Total \# parameters} = \boxed{k p (m-1) + (k-1)}$$

$\downarrow$ likelihood $\downarrow$ prior

Q: 10 features. Each feature can take 5 different values, 3 classes. How many parameters with Naïve Bayes assumption?

Sol. # parameters = $3 \times 10 (5-1) + (3-1)$
 $= 30 \times 4 + 2$
 $= \boxed{122}$

GATE DA'2024

↗ two values

Q.20	Given a dataset with K <u>binary-valued attributes</u> (where $K > 2$) for a <u>two-class classification task</u> , the number of parameters to be estimated for learning a <u>naïve Bayes classifier</u> is
(A)	$2^K + 1$
☒ (B)	$2K + 1$
(C)	$2^{K+1} + 1$
(D)	$K^2 + 1$

dim = K

<u>0/1</u>	<u>0/1</u>	<u>0/1</u>	<u>0/1</u>
x_1	x_2	x_k	y

classes = 2

$P(x_1=0 | y=0)$
 $\downarrow$
 $P(x_1=1 | y=0)$
 $= 1 - P(x_1=0 | y=0)$

$(2-1) \times K \times 2 + (2-1)$
 $\downarrow \qquad \qquad \downarrow$
 likelihood prior

$= \boxed{2K + 1}$

Naive Bayes Algorithm

Bayes Formula :-
$$P(C_k | x) = \frac{P(x | C_k) \cdot P(C_k)}{P(x)}$$

Naive Bayes assumption :-
$$P(x_1, x_2, \dots, x_p | C_k) = P(x_1 | C_k) \cdot \dots \cdot P(x_p | C_k)$$

① Calculate prior probabilities $P(y=C_1), P(y=C_2) \dots P(y=C_k)$
using the dataset.

② Calculate likelihood :- $P(\underbrace{x_1}_{\substack{\text{easy to calculate} \\ \text{in categorical data}}} | y=C_1), P(x_2 | y=C_2) \dots$

③ Calculate posterior probability for all class
 $P(C_1 | x), P(C_2 | x) \dots P(C_k | x)$

④ Assign the label with highest probability

H/W

Candidate	Skill Level	Certification	Previous Job Success	Education Relevance	Hired
1	High	Yes	High	Relevant	Yes
2	High	No	Low	Irrelevant	No
3	Medium	Yes	Medium	Relevant	Yes
4	Low	No	Low	Irrelevant	No
5	Medium	No	Medium	Irrelevant	No
6	Low	Yes	High	Relevant	No
7	High	Yes	Medium	Irrelevant	Yes
8	Medium	Yes	Low	Relevant	No
9	High	No	High	Relevant	Yes
10	Low	No	Medium	Irrelevant	No

Classify whether a new candidate with skill level: Medium, has done certification, Previous Job Success: High, Education Relevant as hired or not?